ITAI 1371 Midterm – Individual Reflection Journals

Team: TheBetaThetas

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Derrick K. Gibbs-McGlaston

What I Learned

Working with the stroke dataset helped me understand why preparing data is an important part of machine learning. Before this project, I knew that missing values and different types of data had to be handled, but seeing these issues in a real dataset made the process easier to understand. I learned that each choice made during preparation can affect the data a model will use later.

The dataset contained 43,400 records. After dividing it into 70% training data and 30% testing data, I focused on the 30,380 training records. I saw that 1,020 training records were missing BMI values and 9,234 were missing smoking status values. Filling missing BMI values with the median and labeling missing smoking status as "unknown" showed me that not every missing value should be handled the same way.

Understanding the Results

The biggest finding for me was how uneven the stroke data was. Before balancing, the training data had 29,832 records without a stroke and only 548 with a stroke. The graph made that difference clear. I learned that a future model could appear accurate simply by predicting that most people did not have a stroke. Balancing the training data gave both groups 29,832 rows, although many stroke rows are repeated examples rather than new people.

I also saw why different columns need different kinds of preparation. One-hot encoding changed categories such as work type and smoking status into numeric columns. Scaling changed the range of age, BMI, and the adjusted glucose values. Creating age groups showed me how information

already in a dataset can be used to make another feature. Comparing the glucose graphs before and after the change helped me see how a data distribution can look different after processing.

Connections and Questions

The most important part of the workflow was splitting the data before exploring and cleaning it. This helped me understand why the testing data should be kept separate while decisions are made using training data. It also connected this project to earlier labs where we trained models and then checked how well they performed on other data.

For the final project, I want to see whether these prepared features help a model identify stroke records, especially because those records are uncommon in the original data. I also understand that balancing the data does not guarantee better predictions. We will need to check more than accuracy, including precision, recall, and the confusion matrix. This project taught me that cleaning data is not just about making the code run; it is about preparing the information carefully so later results can be evaluated fairly.

Victor Djuma

Working on this midterm helped me understand that preparing data is one of the most important parts of machine learning. Before this project, I mostly thought about using data to train a model and make predictions. During this project, I learned that the data has to be carefully inspected, cleaned, and prepared before it can be used correctly.

Our project uses a stroke prediction dataset with 43,400 records and 12 columns. One of the first things I noticed was that the dataset was highly imbalanced. Most of the records were patients without a stroke, while only a small percentage represented stroke cases. This showed me why accuracy alone could be misleading. A model could predict "no stroke" most of the time and still have very high accuracy while failing to identify the patients who actually experienced a stroke.

I also learned why it is important to separate the training and testing data before doing exploratory data analysis and preprocessing. We created a

70% training and 30% testing split and used stratification so that the percentage of stroke cases stayed approximately the same in both sets. The exploratory analysis was then performed using the training data. This helped me understand the importance of preventing information from the test data from influencing the preparation process.

Another important part of this project was dealing with missing data. The BMI and smoking-status columns contained missing values. Instead of simply removing many records, the preprocessing process handled those missing values while keeping useful information. I also learned how categorical data has to be converted into numerical values before most machine learning models can use it.

Feature scaling was another concept that became clearer to me. Some numerical variables have very different ranges and contain extreme values. Using RobustScaler helped prepare these numerical features while reducing the influence of outliers. I also learned that preprocessing should learn its values, such as medians and scaling information, from the training data rather than from the entire dataset.

The class imbalance was probably the most important part for me because it connects directly to the purpose of the project. Stroke cases are rare in this dataset, but those are the cases that would be very important for a prediction model to identify. I learned about balanced class weights and how they can give more importance to the minority stroke class during model training.

I also learned the importance of keeping the original dataset unchanged. Instead of modifying the original CSV directly, the cleaning and preprocessing are performed through Python code in the Jupyter notebook. This makes the work easier to reproduce and allows us to see exactly what changes were made.

Overall, this midterm helped me understand that good machine learning starts with good data preparation. I now have a better understanding of exploratory data analysis, train-test splitting, missing-value handling, categorical encoding, scaling, feature engineering, and class imbalance. This project also prepared me for the final project because the cleaned data and preprocessing steps can later be used when we begin training and evaluating machine learning models.

Pablo Lara

Before working on this project, I understood that machine learning uses data to make predictions, but I did not fully understand how much work goes into preparing that data before training a model. This project helped me understand that the quality of the data and the decisions made during preparation can affect the results.

One thing I learned was the importance of looking at the dataset before trying to use it. Our stroke dataset had 43,400 records, but not all the information was complete. For example, some BMI and smoking-status values were missing. Instead of automatically deleting those records, we looked at different ways to handle the missing information. We decided to fill missing BMI values using the median and mark records that originally had missing BMI. For smoking status, we used "Unknown" instead of guessing a person's smoking status.

Another important thing I learned was how unbalanced data can affect machine learning. Only about 1.8% of the records in our dataset were labeled as stroke cases. I learned that a model could predict "no stroke" for everyone and still have about 98% accuracy. However, it would miss every stroke case. This helped me understand why accuracy alone is not enough and why we need to look at measurements such as precision, recall, F1-score, and PR-AUC when we compare models.

I also learned why we split the data into training and testing sets before doing our analysis. We used a 70/30 split and stratify=y to keep the proportion of stroke cases similar in both sets. I learned that using the test data while making preparation decisions could give us an unfair idea of how well a model performs. Keeping the test data separate helps us evaluate the model more fairly later.

Working through the notebook also helped me understand how Python tools fit together. In Google Colab, I worked through the notebook step by step, reviewed the outputs and graphs, and used explanations to understand what the code was doing. I also worked through errors and questions about the preprocessing pipeline. Using Git Bash and GitHub gave me experience with organizing project files and uploading my work to a repository.

One challenge was understanding the purpose of each preprocessing decision instead of just following the code. For example, I needed to understand why we kept unusual BMI values rather than automatically removing them, why we used one-hot encoding for categories, and why the preprocessing pipeline should be fitted using training data only. Working through these questions helped me connect the code to the reasoning behind it.

Overall, this project changed how I think about machine learning. I now understand that building a model is not just about choosing an algorithm. We also have to inspect the data, handle missing information carefully, prevent data leakage, and choose evaluation measurements that fit the problem. For the final project, I want to build on what I learned by comparing different models and understanding their results. This project is for educational purposes and is not intended to be used as a medical diagnostic tool.

Brandy Gonzales

The hardest thing for me was knowing when to use the mean, median, or mode. Once I worked with the dataset and saw the charts, it made sense visually. The median worked best for BMI because outliers could skew the average.

I learned why we split the data before EDA. We split it in Python into 70% training and 30% testing, and we used stratify so both groups kept the same 1.8% stroke rate. The training set is like study material, and the test set is like a sealed final exam. If the model sees the test data early, its score looks great but it's fake, which is called data leakage. Splitting keeps the test fair.

I also learned the difference between scaling and normalization. Scaling changes the range of the data, and normalization changes the shape. I added a log transform on glucose because it had a long tail of high values, and it made the distribution more balanced.

The most surprising thing was the accuracy trap. Using 5-fold cross-validation, the baseline model scored 98.2% accuracy but 0% recall, because it never caught a single stroke. In healthcare, a missed diagnosis could cost lives, so recall matters more than accuracy here.

This project is for educational purposes only. Next, I will use this cleaned data to train real models and judge them mostly on recall.

Marlon Roberts

Doing this lab, I learned a lot. I was very scared to start it, but once I actually got into it, it was just like all the labs we have been doing since day one. I just had to see what I was working with. Before anything else, you must fill the missing NaN and null values. Scaling was my first step after that, and scaling really puts the data into perspective. Then I normalized it. From there, we balanced the dataset and used one-hot encoding. We also encoded the data, and I learned that one-hot encoding and encoding are two different things. Then we did feature engineering, and boom, you have your first cleaned dataset. You also need a graph to show what the results are. It was a great learning process through it all.